

SKILL SESSION 22&23

Session 22 :

TASK 1: Execute Large Pipelines, Launch Multiple Jobs,
Measure Stability, Monitor Resources, Detect Failures,
Analyze Performance

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    long long sum = 0;
    int i;

    printf("Large pipeline execution started.\n");

    for (i = 1; i <= 10000000; i++)
    {
        sum += i;
    }

    printf("Pipeline calculation completed.\n");
    printf("Sum = %lld\n", sum);

    sleep(5);

    printf("Pipeline execution finished.\n");

    return 0;
}
```

```
#!/bin/bash
echo "==== PIPELINE PERFORMANCE REPORT ====="

echo ""
echo "1. Single Pipeline Execution"

time ./pipeline

echo ""
echo "2. Launching Multiple Jobs"

./pipeline &
./pipeline &
./pipeline &
./pipeline &

echo ""
echo "3. Waiting for All Jobs to Complete"

wait

echo "All background jobs completed."

echo ""
echo "4. Failure Detection Test"

./pipeline_missing

if [ $? -ne 0 ]; then
    echo "Failure detected successfully."
fi

echo ""
echo "==== PERFORMANCE TEST COMPLETED ====="
```

```

-----
(base) saiabburi@Charishma-MacBook-Air ~ % cd Skill12
(base) saiabburi@Charishma-MacBook-Air Skill12 % ls
pipeline      pipeline.c
(base) saiabburi@Charishma-MacBook-Air Skill12 % ./pipeline
Large pipeline execution started.
Pipeline calculation completed.
Sum = 50000005000000
Pipeline execution finished.
(base) saiabburi@Charishma-MacBook-Air Skill12 % time ./pipeline
Large pipeline execution started.
Pipeline calculation completed.
Sum = 50000005000000
Pipeline execution finished.
./pipeline 0.02s user 0.00s system 0% cpu 5.036 total
-----

(base) saiabburi@Charishma-MacBook-Air Skill12 % ./pipeline & ./pipeline & ./pipeline & ./pipeline &
[1] 3538
[2] 3539
[3] 3540
[4] 3541
(base) saiabburi@Charishma-MacBook-Air Skill12 % Large pipeline execution started.
Large pipeline execution started.
Large pipeline execution started.
Large pipeline execution started.
Pipeline calculation completed.
Sum = 50000005000000
Pipeline calculation completed.
Sum = 50000005000000
Pipeline calculation completed.
Sum = 50000005000000
Pipeline calculation completed.
Sum = 50000005000000
Pipeline execution finished.
Pipeline execution finished.
Pipeline execution finished.
Pipeline execution finished.

[4] + done      ./pipeline
(base) saiabburi@Charishma-MacBook-Air Skill12 %
[3] + done      ./pipeline
(base) saiabburi@Charishma-MacBook-Air Skill12 %
[2] + done      ./pipeline
(base) saiabburi@Charishma-MacBook-Air Skill12 %
[1] + done      ./pipeline

```

```
(base) saiabburi@Charishma-MacBook-Air Skill12 % ps -ax | grep pipeline
[ 3624 ttys000      0:00.00 grep pipeline
[(base) saiabburi@Charishma-MacBook-Air Skill12 % ./pipeline_missing
zsh: no such file or directory: ./pipeline_missing
[(base) saiabburi@Charishma-MacBook-Air Skill12 % echo $?
127
[(base) saiabburi@Charishma-MacBook-Air Skill12 % nano performance_report.sh
(base) saiabburi@Charishma-MacBook-Air Skill12 % chmod +x performance_report.sh
[(base) saiabburi@Charishma-MacBook-Air Skill12 % ./performance_report.sh
===== PIPELINE PERFORMANCE REPORT =====
```

```
1. Single Pipeline Execution
Large pipeline execution started.
Pipeline calculation completed.
Sum = 50000005000000
Pipeline execution finished.
```

```
real    0m5.035s
user    0m0.026s
sys     0m0.002s
```

2. Launching Multiple Jobs

```
3. Waiting for All Jobs to Complete
Large pipeline execution started.
Large pipeline execution started.
Large pipeline execution started.
Large pipeline execution started.
Pipeline calculation completed.
Sum = 50000005000000
Pipeline calculation completed.
Sum = 50000005000000
Pipeline calculation completed.
Sum = 50000005000000
Pipeline calculation completed.
Sum = 50000005000000
Pipeline execution finished.
Pipeline execution finished.
Pipeline execution finished.
Pipeline execution finished.
All background jobs completed.
```

4. Failure Detection Test

```
./performance_report.sh: line 28: ./pipeline_missing: No such file or directory
Failure detected successfully.
```

```
===== PERFORMANCE TEST COMPLETED =====
```

```
(base) saiabburi@Charishma-MacBook-Air Skill12 % ./performance_report.sh > performance_report.txt 2>&1
(base) saiabburi@Charishma-MacBook-Air Skill12 % ls
performance_report.sh  pipeline
performance_report.txt pipeline.c
```

